Lecture 7

Reviews

Faculty of Information Technology
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Outline

- Review objectives
- Formal design reviews (DRs)
- Peer reviews

References

- Daniel Galin, Software Quality Assurance: From theory to implementation, chapter 8

Review objectives

Why Review?

The Problem:

- Design documents are typically created by System Analysts
- Authors and team leaders are often unable to detect their own errors due to over-familiarity with the document

The Solution:

- Involve external parties who were not directly involved in creating the document
- **Participants:** Peers, superiors, external experts, and customer representatives

The Goal: Utilization of different viewpoints and experiences to detect unnoticed errors

Review

"A process or meeting during which a work product... is presented to project personnel, managers, users, customers, or other interested parties for comment or approval."

Importance in SQA:

- **Early Detection:** Identifies defects during the initial analysis and design phases
- **Cost Reduction:** Prevents errors from passing "downstream" to coding/testing stages, where detection and correction are much more intricate and costly

Review methodologies

Several methodologies can be implemented:

- **Formal Design Reviews (DRs)**
- **Peer Reviews:**
  - Inspections
  - Walkthroughs
- **Expert Opinions**

Strategy: Apply a "double or triple net"

Direct Objectives of Reviews

Deal directly with the **current project:**

- **Error Detection:** Identify analysis and design defects vs specifications
- **Risk Identification:** Spot issues affecting project completion
- **Standardization:** Locate deviations from templates and style standards
- **Approval:** Authorize the product for the next phase

Indirect Objectives of Reviews

Deal directly with **professional growth and process improvement:**

- **Knowledge Exchange:** Share professional methods, tools, and techniques among the team
- **Process Improvement:** Record errors to drive future **corrective actions** and quality enhancements

Principles for Effective Reviews

Structured Approach: Must follow procedural order, not haphazard

Key Roles:

- Coordinator: Keeps the meeting on track
- Scribe: Records agreed-upon defects in detail

The Golden Rule:

- Detect errors only
- Do not design solutions on the spot

Formal design reviews

Formal design reviews (DRs)

- Also known as “Formal Technical Reviews” (FTR)
- Without DR approval, the development team **cannot** continue to the next phase of the project
- **Conducted at any development milestone** requiring the completion of an analysis or design document (e.g., Requirement Specification, Installation Plan)

Formal design reviews (DRs)

DPR – Development Plan Review
SRSR – Software Requirement Specification Review
PDR – Preliminary Design Review
DDR – Detailed Design Review
DBDR – Data Base Design Review
TPR – Test Plan Review
STPR – Software Test Procedure Review
VDR – Version Description Review
OMR – Operator Manual Review
SMR – Support Manual Review
TRR – Test Readiness Review
PRR – Product Release Review
IPR – Installation Plan Review

Some common formal design reviews

Participants - The review leader

- **Importance:** The appointment of the leader is a major factor in the DR's success
- **Required Characteristics:**
  - **Knowledge:** Experience in projects of the type being reviewed
  - **Seniority:** Level similar to or higher than the Project Leader
  - **Position:** Must be external to the project team (e.g., QA Manager, Chief Software Engineer)
  - **Relationship:** Good rapport with the project leader and team

Participants - The review team

- **Composition:**
  - Senior members of the project team
  - Senior professionals from other projects/departments
  - Customer/User representatives
  - **Best Practice:** Non-project staff should make up the majority
- **Optimal Size:**
  - 3 to 5 members
  - **Warning:** Excessively large teams create coordination problems and waste time

Preparation for a DR

Review Leader:

- Appoints the team, schedules the session, and distributes documents (hard copy/electronic)
- Timing: Schedule shortly after distribution to prevent delays

Review Team:

- Must review the document and list comments prior to the session
- Use **Checklists** to ensure no issues are overlooked

Development Team:

- Prepares a short presentation focusing on main professional issues, not a general project description

The DR session agenda

1. **Presentation:** Short overview of the design document
2. **Discussion:** Comments made by the review team
3. **Verification:** Each comment is discussed to determine required corrections
4. **Decision:** The team determines the project's progress status

DR Decision outputs

The review team makes one of three decisions:

- **Full Approval:** Immediate continuation to the next phase (minor corrections may be requested).
- **Partial Approval:**
  - Approved parts proceed.
  - Non-approved parts require major corrections and must be reviewed again before proceeding.
- **Denial of Approval:** Demands a repeat of the DR due to multiple major/critical defects

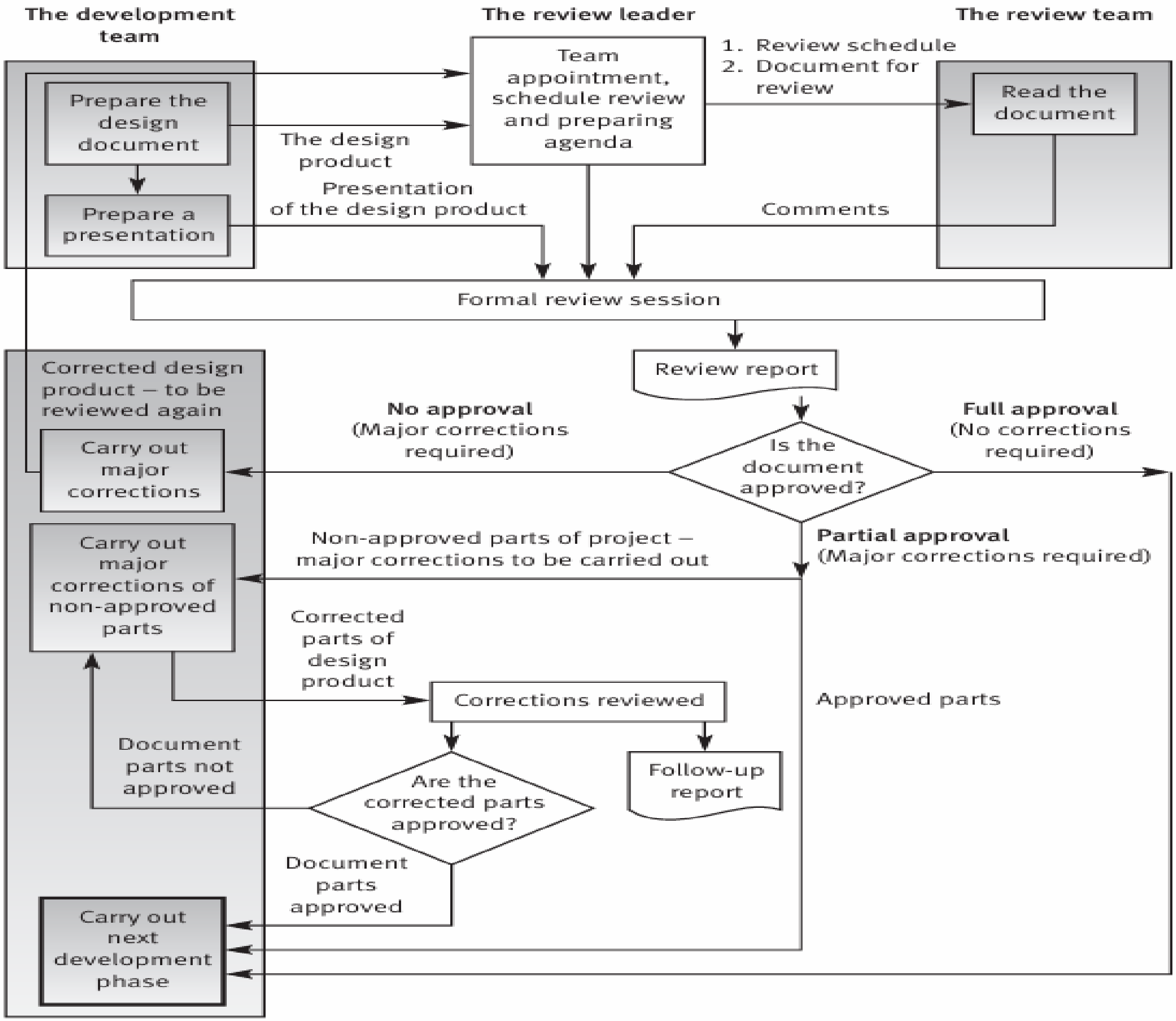
Post-review activities

- **The DR Report:**
  - Issued immediately by the Review Leader
  - Contains: Summary of discussion, Decision, Full list of Action Items (who/when), and Follow-up assignments
- **The Follow-up Process:**
  - The appointed person checks if action items are satisfactorily accomplished
  - This is a condition for allowing the project to continue

Pressman's "Golden Guidelines"

- **Infrastructure:** Develop checklists, train senior professionals, schedule DRs in the project plan
- **The Session:**
  - Limit team size (**3-5** members)
  - Keep the **meeting professional** (no personal attacks)
  - **Stick** to the agenda
  - Limit duration: **Max 2 hours**
  - Rule: Focus on **detecting** defects, not discussing solutions

The formal design review process



Peer reviews

Peer reviews

- **Participants: Equals (peers)** to the project leader/author
- **Authority:** Unlike DRs, they are **not authorized** to approve the design document for the next stage
- **Objective:** Detecting errors and deviations from standards
- **Effectiveness:** Highly efficient for SQA

Inspection vs. Walkthrough

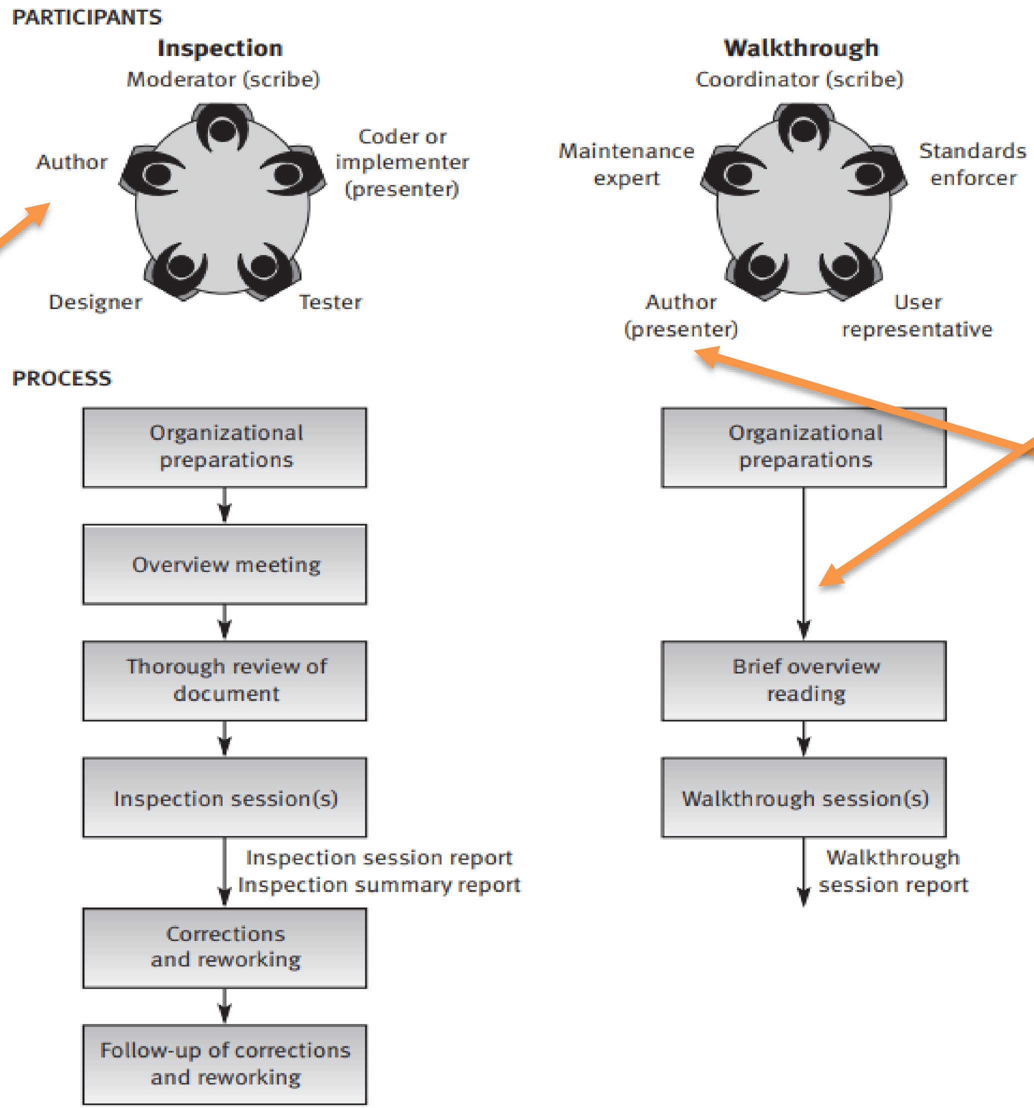
Inspection:

- Higher level of formality
- Objective includes corrective action and process improvement (SQA contribution)

Walkthrough:

- Lower level of formality
- Findings are usually limited to comments on document

Inspection vs. Walkthrough



Author is **not** the presenter

Much less formal…

Author is the presenter

Peer reviews participants

- **Team structure:**
  - Optimal size is 3 to 5 participants
  - **Review leader:** “Moderator” (Inspection) or “Coordinator” (Walkthrough). Come from outside the project team, experience in projects of the type being reviewed
  - **The author:** Always participates
- **Specialized roles:**
  - Inspection: Designer, Coder and Tester
  - Walkthrough: Standards Enforcer, Maintenance Expert and User Representative

Key roles: Presenter & Scribe

- **The Presenter:**
  - Inspection: Usually **not the author** (often the Coder). This tests if the design logic is clear to someone else
  - Walkthrough: Usually **the Author**, as they are most familiar with the document
- **The Scribe:**
  - Records all discussed defects
  - This role requires professional understanding of the technical issues (not just administrative work)

Peer review leader’s preparations

- **Define Scope:** Select critical, complex, or defect-prone sections with the author
- **Select Team:** Appoint appropriate members
- **Scheduling:**
  - Max **2 hours** per session
  - Schedule up to two sessions a day if review tasks is sizable
  - For Inspections: Schedule an overview meeting for the team
- **Distribution:** Distribute the document to team members prior to the session

Peer review team's preparations

- **Inspection Team (Thorough):**
  - Mandatory: Read & list comments before the session
  - Context: Attend the Overview Meeting
  - Tool: Use **Checklists** to catch all issues
- **Walkthrough Team (Brief):**
  - Overview: Brief reading to understand the general concept
  - Requirement: Usually, no advance **preparation** of comments is needed

The Session & Documentation

- **The Session:**
  - Review Leader guides the flow
  - Discussion confined to identifying errors, not solutions
  - Max duration: 2 hours
- **Documentation:**
  - Inspection: Scribe records defect location, description, and Severity (Critical/Major/Minor) for statistical analysis
  - Walkthrough: Produces a "findings report" (list of errors)

Post-peer review activities

- The most fundamental differentiating element between the two peer review methods
- **Inspection (Comprehensive):**
  - **Does not end** with a review session or distribution of reports
  - Requires follow-up to ensure errors are fixed
  - Reports go to the Corrective Action Board (CAB) to analyze trends and improve development methods

Coverage and Efficiency

- **Coverage:**
  - 5–15% of documents usually undergo peer review
  - Strategy: Focus on high-risk, complex, or defect-prone sections
- **Efficiency Metrics:**
  - Average hours worked per defect detected
  - Defect detection density (defects per page)
  - Internal effectiveness (% of defects found by review vs. testing)

Summary

- Review objectives
- Formal design reviews (DRs)
- Peer reviews